

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 12
 - Limits of sequences
 - Cauchy sequences

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Metric space version:

$\forall \epsilon > 0$, there exists $N \in \mathbb{Z}_+$ such that $d(x_n, L) < \epsilon \forall n \geq N$.

We say $\{x_n\}$ **converges** to L and L is the **limit of the sequence**.

Example

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Take $\epsilon = 1$ and choose $N \in \mathbb{Z}_+$ such that $d(x_n, L) < 1$ for all $n \geq N$.

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Take $\epsilon = 1$ and choose $N \in \mathbb{Z}_+$ such that $d(x_n, L) < 1$ for all $n \geq N$.

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This means $x_n = 1/n = L$ for all $n \geq N$, which is a contradiction! □

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Then there exists N_1 such that $d(x_n, L) < \epsilon$ for $n \geq N_1$.

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Then there exists N_1 such that $d(x_n, L) < \epsilon$ for $n \geq N_1$.

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This is a contradiction.



Existence of limits

Theorem

If a sequence $\{x_n\}$ in a metric space (M, d) converges to a value $L \in M$, then the range

$$X = \{x_1, x_2, \dots\} \subseteq M$$

is a bounded set and L is an adherent point of X .

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When the range of a sequence is bounded, we call the sequence $\{x_n\}$ bounded.

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If we take

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then $d(x_n, L) \leq R$ for all N .

Therefore $X \subseteq B_M(L, R)$ and in particular X is bounded. □

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Theorem

If S is a subset of a metric space (M, d) and $L \in M$ is an adherent point of S , then there exists a sequence $\{x_n\}$ of elements of S which converges to L .

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We claim $\lim x_n = L$.

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Since $\epsilon > 0$ was arbitrary, this proves convergence. □

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Theorem

A sequence $\{x_n\}$ in a metric space (M, d) converges to a value $L \in M$ if and only if every subsequence $\{x_{k(n)}\}$ also converges to L .

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To show it limits to L , let $\epsilon > 0$.

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To show it limits to L , let $\epsilon > 0$.

Then there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d(x_n, L) < \epsilon$.

Moreover, $k(n) \geq n$ so $k(n) \geq N$ and therefore $d(x_{k(n)}, L) < \epsilon$.

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Since $\epsilon > 0$ was arbitrary, this proves convergence. $\square$

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In any of these cases, we call the sequence **monotone**.

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A monotone decreasing sequence $\{x_n\}$ which is bounded below converges to $L = \inf\{x_1, x_2, \dots\}$.

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Suppose that $\{x_n\}$ is a monotone increasing sequence which is bounded above and let $X = \{x_1, x_2, \dots\}$.

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Since $\epsilon > 0$ was arbitrary, this proves x_n converges to L . □

Outline

- 1 Real Analysis Lecture 12
 - Limits of sequences
 - Cauchy sequences

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Hence $X = \{x_1, x_2, \dots\} \subseteq B_M(x_N, R)$, and is bounded.